

CHAPTER 2: SURVEY OF TECHNOLOGIES

2.1 Introduction

The Smart Event Management and Ticket Booking System is developed using modern web technologies that provide a secure, scalable, and user-friendly environment for managing events. The selected technologies ensure efficient communication between the user interface, server, and database while supporting responsive design and fast performance. This chapter provides an overview of the technologies used in the development of the proposed system and explains their significance.

2.2 Front-End Technologies

2.2.1 React.js

React.js is an open-source JavaScript library developed by Facebook for building interactive and dynamic user interfaces. It follows a component-based architecture, allowing developers to create reusable UI components that improve code maintainability and application performance. React uses a Virtual DOM to efficiently update only the necessary parts of a webpage, resulting in faster rendering and a better user experience.

Advantages of React.js

- Component-based architecture
 - High performance using Virtual DOM
 - Reusable code
 - Easy state management
 - Responsive user interface
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2.2.2 HTML5

HTML5 (HyperText Markup Language) is the standard markup language used for designing the structure of web pages. It provides semantic elements that improve accessibility and maintainability while supporting multimedia content without external plugins.

Features

- Semantic tags
 - Audio and video support
 - Cross-platform compatibility
 - Improved form validation
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2.2.3 CSS3

CSS3 (Cascading Style Sheets) is used to design and style web pages. It provides responsive layouts, animations, transitions, gradients, and modern UI components that improve the overall appearance of the application.

Features

- Responsive design
 - Animations
 - Flexbox and Grid layouts
 - Media queries
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2.2.4 Bootstrap 5

Bootstrap is a front-end framework used to develop responsive and mobile-friendly web applications quickly. It contains pre-built components such as navigation bars, cards, forms, tables, buttons, and modals.

Advantages

- Faster UI development
 - Mobile-first design
 - Responsive layouts
 - Consistent styling
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2.3 Back-End Technologies

2.3.1 Node.js

Node.js is a server-side JavaScript runtime environment used to execute JavaScript outside the browser. It is lightweight, efficient, and suitable for building scalable web applications with asynchronous request handling.

Advantages

- Fast execution
 - Non-blocking architecture
 - Scalable applications
 - JavaScript used for both frontend and backend
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2.3.2 Express.js

Express.js is a lightweight web application framework built on Node.js. It simplifies server-side development by providing routing, middleware support, request handling, and RESTful API development.

Advantages

- Simple routing
 - Middleware support
 - Easy API development
 - Lightweight and flexible
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2.4 Database Technology

2.4.1 MongoDB Atlas

MongoDB Atlas is a cloud-based NoSQL database used for storing application data such as users, events, bookings, payments, and feedback. It stores data in JSON-like documents, making it flexible and scalable for modern web applications.

Advantages

- Cloud hosted
 - Flexible document-based structure
 - High scalability
 - Secure authentication
 - Automatic backup
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2.5 Development Tools

2.5.1 Visual Studio Code

Visual Studio Code is a lightweight source-code editor used for developing the application. It provides features such as IntelliSense, debugging, integrated terminal, Git integration, and extension support.

2.5.2 Git

Git is a distributed version control system used to track changes in source code during software development. It allows developers to maintain version history and collaborate efficiently.

2.5.3 GitHub

GitHub is a cloud-based repository hosting platform that stores Git repositories online. It provides version control, collaboration features, issue tracking, and project management capabilities.

2.6 Summary

The proposed Smart Event Management and Ticket Booking System utilizes a modern technology stack consisting of React.js for the frontend, Node.js and Express.js for backend development, and MongoDB Atlas for database management. These technologies provide a secure, scalable, and efficient platform capable of handling event management operations while delivering a responsive and user-friendly experience. The selected tools and frameworks also simplify development, maintenance, and future enhancements of the system.